Kazi Arman Ahmed

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Stillwater, Oklahoma, USA

Research Interests

Stochastic and decentralized optimization; privacy-preserving coordination of energy systems under uncertainty; federated learning; agentic AI for optimization; healthcare analytics.

Education

Oklahoma State University Aug 2025 – May 2027 (expected)
M.S. in Business Analytics and Data Science Stillwater, OK

- **GPA:** 4.00/4.00. Graduate Certificate in Business Analytics and Data Science awarded Dec 2025.

Oklahoma State University Aug 2024 – Jul 2026
M.S. in Industrial Engineering and Management (Thesis) Stillwater, OK

- **GPA:** 4.00/4.00. **Concentration:** Operations Research and Analytics. **Advisor:** Dr. Paritosh Ramanan.
- **Thesis:** Privacy-Preserving Decentralized Stochastic Reserve Coordination of Electrified Chemical Plants with Renewable-Driven Transmission System [M1].
- **Coursework:** Introduction to Optimization, Probability and Statistics, Discrete Systems Simulation, Decentralized Blockchain Computing for Engineering Applications, Predictive Business Analytics, Advanced Data Wrangling.

Military Institute of Science and Technology (MIST) Feb 2016 – Jan 2020
B.Sc. in Industrial and Production Engineering Dhaka, Bangladesh

- **GPA:** 3.58/4.00. Merit Scholarship (top 5 of the academic batch, 2017 and 2018).
- **Thesis:** Statistical Optimization of Selective Laser Sintering for Thermoplastic Polyurethane [J2].

Publications

Own name in bold.

Manuscripts Under Review and in Preparation

[M1] **K. A. Ahmed**, H. Q. N. Vo, S. Ghasemi Naraghi, P. Ramanan, and Z. Jiang. "Privacy-Preserving Decentralized Stochastic Reserve Coordination of Electrified Chemical Plants with Renewable-Driven Transmission Systems." [Manuscript in preparation for *IEEE Transactions on Power Systems*, 2026]

[M2] **K. A. Ahmed***, R. Reed*, S. Ghasemi, Z. Jiang, and P. Ramanan. "Decentralized Operations of Decarbonized Chemical Plants with Renewable-Driven Transmission Systems." [Submitted to *Electric Power Systems Research*, 2026] arXiv:2606.22973. *Equal contribution.

[M3] **K. A. Ahmed**, K. F. Anjum, and I. Humaira. "Personalized Federated Load Forecasting for Distributed Industrial Power Systems with Cold-Start Transfer to a New Chemical Plant." [Submitted to IEEE Texas Power and Energy Conference (TPEC), 2027]

[M4] C. Deshpande and **K. A. Ahmed**. "Decentralized, Privacy-Preserving Early Warning of Acute Kidney Injury in Intensive Care: A Decision-Oriented Benchmark." [Submitted to Southwest Decision Sciences Institute (SWDSI) Annual Meeting, 2027]

Journal Articles

[J1] **K. A. Ahmed**, I. Humaira, A. R. Khan, M. S. Hasan, M. Islam, A. Roy, M. Karim, M. Uddin, A. Mohammad, and M. D. Xames. "Advancing Breast Cancer Prediction: Comparative Analysis of ML Models and Deep Learning-Based Multi-Model Ensembles on Original and Synthetic Datasets." *PLOS ONE*, 20(6):e0326221, 2025.

[J2] M. M. Rahman, **K. A. Ahmed**, M. Karim, J. Hassan, R. Roy, B. Bustami, S. M. N. Alam, and H. Younes. "Optimization of Selective Laser Sintering Three-Dimensional Printing of Thermoplastic Polyurethane Elastomer: A Statistical Approach." *Journal of Manufacturing and Materials Processing*, 7(4):144, 2023.

[J3] A. Roy, M. Islam, M. Karim, **K. A. Ahmed**, A. R. Khan, M. Uddin, and M. D. Xames. "Comparative Analysis of KNN and SVM in Multicriteria Inventory Classification Using TOPSIS." *International Journal of Information Technology*, 15(7):3613–3622, 2023.

Conference Proceedings

[C1] **K. A. Ahmed**, H. Hadidi, M. Fili, D. Delen, G. Hu, and L. Wang. "Comorbidity Patterns Before and After Juvenile Idiopathic Arthritis Diagnosis." In *AI, Society and Digital Transformation: Proceedings of the 2025 INFORMS Conference on Service Science*, University of Oxford, UK. Lecture Notes in Operations Research, Springer, pp. 276–288, 2026.

[C2] R. Rahman, **K. A. Ahmed**, M. Fili, H. Hadidi, G. Hu, D. Delen, and R. B. Azad. "Post-Diagnosis Comorbidity Patterns in Patients with Still's Disease." In *IISE Annual Conference Proceedings*, Institute of Industrial and Systems Engineers (IISE), pp. 1–6, 2026.

Research Experience

Graduate Research Assistant 2025 – 2026
Distributed Intelligent Systems (DISys) Laboratory, Oklahoma State University Advisor: Dr. Paritosh Ramanan

- Formulated privacy-preserving reserve coordination between a 2,000-bus transmission grid (ACTIVSg2000) and 26 electrified ethane-cracker microgrids, each modeled as a two-stage stochastic program with non-anticipative reserve bids [M1].
- Designed a two-phase, two-penalty ADMM that couples grid and plants through reserve capacity rather than power demand; implemented it as an MPI program on HPC clusters with Gurobi.
- Matched the centralized benchmark to within 1.3% in cost across 18 experiments and within 0.7% in system emissions in every comparison, while plants shared only reserve bids.
- Evaluated the framework under three renewable scenario generators, including a labmate's importance-weighted split variational autoencoder that generates scenarios without pooling plant data.
- Developed, as co-first author, a deterministic decentralized ADMM framework for joint day-ahead scheduling of the grid and 26 chemical plants, with costs within 2% of the centralized benchmark [M2].
- Built personalized federated forecasters across heterogeneous manufacturing plants and a cold-start transfer method for a newly connected chemical plant with about one week of history, without pooling any plant's data [M3].
- ***In progress:*** Developing agent skills for the lab's multi-agent framework, **Scarlet Composer**, in which the grid and each plant solve their own subproblems as agents and an LLM coordinator adapts the ADMM penalty from privacy-preserving residual signals.

Healthcare Analytics Research 2024 – 2025
Healthcare Data Analytics Lab, George Mason University With Drs. G. Hu, M. Fili, L. Wang, D. Delen

- Built cohort-construction and comorbidity-analysis pipelines on Oracle Real-World Data, a multi-system clinical database covering millions of patients, to characterize comorbidity patterns before and after diagnosis of Juvenile Idiopathic Arthritis (9,257 patients) and Still's disease [C1], [C2].

Industry Experience

Advanced Analytics Intern May 2026 – Present
Data Research and Analytics, AAA, The Auto Club Group Remote

- Build conversational data agents with Google ADK, Gemini, and the Conversational Analytics API that answer natural-language questions over BigQuery tables and Tableau dashboards.
- Built a multi-agent evaluation framework that generates reference questions and validates agent answers against BigQuery across 100 queries to detect hallucinations before deployment.

Data Analyst Dec 2020 – Jul 2024
Data Management Department, LandQuire Management LLC Dhaka, Bangladesh

- Automated ETL pipelines and dashboards with Python, SQL, and Supermetrics APIs, reducing manual reporting workload by 85% (Certificate of Excellence).
- Built a property acquisition data pipeline with cross-functional teams, expanding data coverage by 40%; trained junior analysts, raising team productivity by 50%.

Teaching Experience

Graduate Teaching Assistant, ENGR 1412 Aug 2024 – Dec 2024
Oklahoma State University (Instructor: Dr. Jennifer Glenn) Stillwater, OK

- Led Python and Excel labs and office hours for 60+ students; about 95% positive student evaluations.

Lecturer, Industrial and Production Engineering Jun 2020 – Dec 2020
European University of Bangladesh Dhaka, Bangladesh

- Taught operations research, manufacturing, and optimization courses; mentored 30+ students on capstone projects.

Presentations

IISE Annual Conference and Expo (three presentations), Arlington, TX 2026
INFORMS Annual Meeting, Atlanta, GA 2025
2nd Annual IEM Research Symposium, Oklahoma State University (Token Prize) 2025

Honors and Awards

National Champion, Data4Good Competition, Johns Hopkins University 2026
Finalist (Top 5), IISE Energy Analytics Challenge 2026
1st Runner-Up, OSU Datathon, Spears School of Business 2026
Winner, M.S. BAnDS Business Case Competition, Oklahoma State University 2026
Creativity, Innovation and Entrepreneurship (CIE) Scholar, Oklahoma State University 2026
Alpha Pi Mu, Industrial Engineering Honor Society 2026
SAS Advisory Board Scholarship, Oklahoma State University 2025
Winner, AlphaRamos Marketing Hackathon 2025
Research Symposium Token Prize, IEM Research Symposium, Oklahoma State University 2025
Tau Beta Pi, Engineering Honor Society 2025
Certificate of Excellence, LandQuire Management LLC 2024
Merit Scholarship, Military Institute of Science and Technology 2017, 2018

Professional Service and Leadership

Journal Reviewer: *Expert Systems with Applications*; *Healthcare Analytics* (Elsevier) 2024 – Present
GPSGA Representative, Business Analytics Student Association, Oklahoma State University 2025 – Present
Vice President, INFORMS Student Chapter, Oklahoma State University 2024 – 2025
Memberships: INFORMS, IISE, Tau Beta Pi, Alpha Pi Mu

Technical Skills

Optimization	Gurobi, Pyomo, AMPL, CPLEX
Programming	Python, R, SQL, MATLAB, C, JavaScript
ML and AI	Supervised & Unsupervised Models, Federated Learning, Agentic AI
Data and Cloud	Google Cloud (BigQuery, Dataplex), Azure, Apache Spark, Tableau, Power BI
Other	Hyperledger Besu, IPFS, SIMIO, SAS, JMP

References

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Industrial Engineering and Management
Oklahoma State University
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Dr. Zheyu Jiang
Assistant Professor
Chemical Engineering
Oklahoma State University
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Dr. Jennifer Glenn
Associate Professor of Teaching
Industrial Engineering and Management
Oklahoma State University
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